

CSE-214 Online

Behavioural Design Pattern

Duration: 25 Minutes

Problem: Automatic AI Model Selection System

An AI platform provides access to three language models with different levels of capability, computational cost, and usage limits. The available models are:

- FlashMind - Fast Model
- CoreMind - Balanced Model
- DeepMindX - Advanced Model

Whenever a user submits a prompt, the system analyzes the complexity of the prompt and automatically selects an appropriate model to generate the response.

Available AI Models

FlashMind

FlashMind is designed for simple requests.

Capability : Basic
Processing Cost : Low
Prompt Limit : Unlimited

CoreMind

CoreMind is designed for moderately complex requests.

Capability : Medium
Processing Cost : Medium
Prompt Limit : 50 prompts

DeepMindX

DeepMindX is designed for difficult requests requiring advanced reasoning.

Capability : High
Processing Cost : High
Prompt Limit : 10 prompts

All three models provide the same general operation:

```
generateResponse(String prompt)
```

However, each model may process the prompt differently.

Prompt Complexity Analysis

Before selecting a model, the system determines the complexity of the submitted prompt.

Assume that a ComplexityAnalyzer component is already available.

It accepts a prompt and returns a random complexity score between 0 and 100.

Model Selection

The required model is determined from the complexity score.

Complexity 0 - 30
-> FlashMind

Complexity 31 - 70
-> CoreMind

Complexity 71 - 100
-> DeepMindX

Model Usage Limits

The platform restricts the number of prompts processed by the more expensive models.

Model	Maximum Number of Prompts
FlashMind	Unlimited
CoreMind	50
DeepMindX	10

The system must keep track of how many prompts have already been processed by each model.

For example:

DeepMindX Used = 10 / 10

CoreMind Used = 31 / 50

In this case, DeepMindX cannot process another prompt.

Fallback Rule

If the selected model has reached its usage limit, the system must try the next lower model.

The fallback order is:

DeepMindX

|

v

CoreMind

|

v

FlashMind

Example Scenario 1

Assume:

Complexity Score = 84

Maximum Allowed Model = DeepMindX

DeepMindX Usage = 7 / 10

CoreMind Usage = 24 / 50

The result is:

Required Model = DeepMindX

Selected Model = DeepMindX

After processing the prompt:

DeepMindX Usage = 8 / 10

Example Scenario 2

Assume:

Complexity Score = 90

Maximum Allowed Model = DeepMindX

DeepMindX Usage = 10 / 10

CoreMind Usage = 37 / 50

The result is:

Required Model = DeepMindX

DeepMindX unavailable due to usage limit.

Selected Model = CoreMind

Example Scenario 3

Assume:

Complexity Score = 22

Maximum Allowed Model = DeepMindX

The prompt is simple, therefore:

Required Model = FlashMind

Selected Model = FlashMind

A more powerful model should not be selected unnecessarily.

You may introduce any necessary classes, interfaces, enumerations, and methods to produce a clean object-oriented solution.